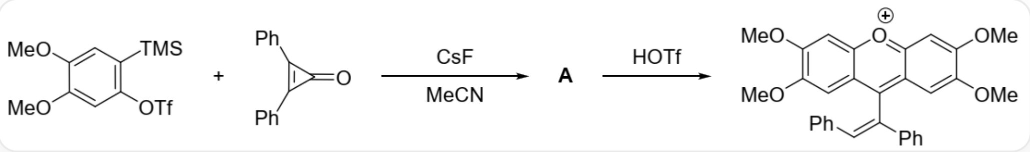
Question

Diphenylcyclopropenone can undergo the following reaction:



`COC1=CC(OS(=O)(C(F)(F)F)=O)=C([Si](C)(C)C)C=C1OC` and `O=C1C(C2=CC=CC=C2)=C1C3=CC=CC=C3` react with cesium fluoride in acetonitrile to generate $\mathbf{A}$, and $\mathbf{A}$ generates `COC1=C(OC)C=C(C(/C(C2=CC=CC=C2)=C\C3=CC=CC=C3)=C(C=C(OC)C(OC)=C4)C4=[O+]5)C5=C1` under the action of $TfOH$

Examine the structural formula of $\mathbf{A}$ and the structures of the three key intermediates $\mathbf{B1}$, $\mathbf{B2}$, and $\mathbf{B3}$ for the generation of $\mathbf{A}$.

The following statements are made:

1. $\mathbf{A}$ contains a three-membered ring.

2. $\mathbf{B1}$ contains a benzyne structure.

3. $\mathbf{B2}$ has five rings.

4. The number of aromatic rings in $\mathbf{B3}$ is the same as that in $\mathbf{B2}$.

Select the option below that contains all the correct statements:

A. All other options are incorrect

B. 1

C. 2

D. 3

E. 4

F. 1,2

G. 1,3

H. 1,4

I. 2,3

J. 2,4

K. 3,4

L. 1,2,3

M. 1,2,4

N. 1,3,4

O. 2,3,4

P. 1,2,3,4

Answer

Correct Answer: $\mathbf{L}$

Detailed Explanation

The product is a positive ion, which is easily understood to be produced by protonation with $TfOH$. Observing the product structure, a hydrogen appears on the sp^2 carbon where the exocyclic double bond is bonded to a phenyl group, which should be generated by protonation. Therefore, $\mathbf{A}$: `COC1=C(OC)C=C(C2(C(C3=CC=CC=C3)=C2C4=CC=CC=C4)C(C=C(OC)C(OC)=C5)=C5O6)C6=C1`

CHECKPOINT 1 PTS

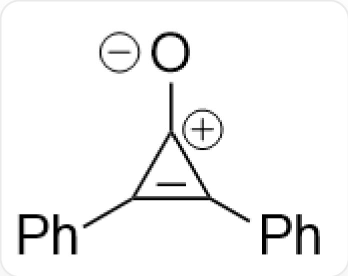
The structure of $\mathbf{A}$ is: `COC1=C(OC)C=C(C2(C(C3=CC=CC=C3)=C2C4=CC=CC=C4)C(C=C(OC)C(OC)=C5)=C5O6)C6=C1`, containing a three-membered ring, statement 1 is correct

To study the process of generating $\mathbf{A}$, cesium fluoride first removes the silyl group from the raw material `COC1=CC(OS(=O)(C(F)(F)F)=O)=C([Si](C)(C)C)C=C1OC` to generate a negative ion, and then TfO^- is removed to generate benzyne, i.e., $\mathbf{B1}$: `COC1=CC#CC=C1OC`

CHECKPOINT 1 PTS

The structure of $\mathbf{B1}$ is `COC1=CC#CC=C1OC`, containing benzyne, statement 2 is correct

The diphenylcyclopropenone of the raw material can be considered to have a charge-separated structure:



`[O-][C+]1C(C2=CC=CC=C2)=C1C3=CC=CC=C3`

It reacts with the generated benzyne to form a ring to obtain **B2**: `COC1=CC2=C(C3(C(C4=CC=CC=C4)=C3C5=CC=CC=C5)O2)C=C1OC`.

CHECKPOINT 1 PTS

The structure of **B2** is `COC1=CC2=C(C3(C(C4=CC=CC=C4)=C3C5=CC=CC=C5)O2)C=C1OC`, with 5 rings, statement 3 is correct

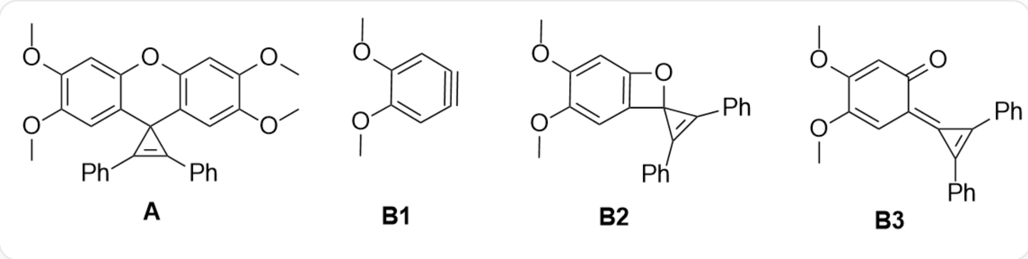
Then the unstable benzocyclobutene ring opens to obtain **B3**: `COC(C(OC)=C/C1=C2C(C3=CC=CC=C3)=C\2C4=CC=CC=C4)=CC1=O`

CHECKPOINT 1 PTS

The structure of **B3** is `COC(C(OC)=C/C1=C2C(C3=CC=CC=C3)=C\2C4=CC=CC=C4)=CC1=O`, which has two benzene rings and two aromatic rings, one less than in **B2**, statement 4 is incorrect

Then **B3** undergoes a Diels-Alder reaction with another molecule of **B1** to obtain **A**.

Statements 1, 2, and 3 are correct, choose L



A：`COC1=C(OC)C=C(C2(C(C3=CC=CC=C3)=C2C4=CC=CC=C4)C(C=C(OC)C(OC)=C5)=C5O6)C6=C1`；**B1**：`COC1=CC#CC=C1OC`；**B2**：`COC1=CC2=C(C3(C(C4=CC=CC=C4)=C3C5=CC=CC=C5)O2)C=C1OC`；**B3**：`COC(C(OC)=C/C1=C2C(C3=CC=CC=C3)=C\2C4=CC=CC=C4)=CC1=O`